



Where is the human in human-centered AI? Insights from developer priorities and user experiences

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ABSTRACT

Human-centered artificial intelligence (HCAI) seeks to shift the focus in AI development from technology to people. However, it is not clear whether existing HCAI principles and practices adequately accomplish this goal. To explore whether HCAI is sufficiently focused on people, we conducted a qualitative survey of AI developers ($N = 75$) and users ($N = 130$) and performed a thematic content analysis on their responses to gain insight into their differing priorities and experiences. Through this, we were able to compare HCAI in principle (guidelines and frameworks) and practice (developer priorities) with user experiences. We found that the social impact of AI was a defining feature of positive user experiences, but this was less of a priority for developers. Furthermore, our results indicated that improving AI functionality from the perspective of the user is an important part of making it human-centered. Indeed, users were more concerned about being understood by AI than about understanding AI. In line with HCAI guidelines, developers were concerned with issues such as ethics, privacy, and security, demonstrating an ‘avoidance of harm’ perspective. However, our results suggest that an increased focus on what people need in their lives is required for HCAI to be truly human-centered.

1. Introduction

Systems incorporating artificial intelligence (AI) have become an inseparable part of many people’s lives, in ways that have positive and negative consequences for users, other people, and society (Oppermann et al., 2019). AI systems are used in a broad range of applications, as varied as connecting people (Willis, 2018), providing entertainment (2017Plummer), and helping to create and distribute vaccines (2021Greig). However, embedding AI in systems can have profound negative consequences (Burrell & Fourcade, 2021; Scheuerman et al., 2020)—as seen when people are incorrectly denied unemployment benefits on the basis of facial recognition software (2021Svaldi), fired by a performance algorithm because of circumstances out of their control (2021Soper), or have their personal information leaked by a chatbot (Jang, 2021). Reflecting the increasing importance and integration of AI in people’s lives, there is a move towards human-centered AI (HCAI), which has the putative goal of placing the user rather than technology at the center of AI development (Shneiderman, 2020a; 2020b). However,

HCAI guidelines tend to focus on broad concepts such as human values, ethics, and privacy, which can be overly abstract and thus difficult to implement in practice (Mittelstadt, 2019; Shneiderman, 2021). Moreover, it is unclear to what extent these guidelines—and the practices that should follow from attempting to apply them—are truly ‘human-centered’ in the sense of focusing on how AI affects people. With the aim of exploring in what sense and to what extent HCAI is human-centered, in this paper we seek to compare HCAI in principle (i.e., as set out in existing guidelines) and in practice (i.e., as evidenced by self-reported developer priorities) with the experiences of AI users.

1.1. HCAI theories, principles, and practices

As described by Shneiderman (2020a, 2020b), the drive of HCAI to put humans at the heart of AI represents a ‘second Copernican revolution’. What this means specifically is a matter of some debate. For Shneiderman (2020b, p. 2), HCAI focuses on human experiences, satisfaction, and needs, with the aim of “amplifying, augmenting, and

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enhancing human performance in ways that make systems reliable, safe, and trustworthy” so as to “support human self-efficacy, encourage creativity, clarify responsibility, and facilitate social participation”. However, other HCAI researchers construe the meaning of ‘human-centered’ in different ways (Kaluvarachchi et al., 2021; Wilkens et al., 2021). For example, Gillies et al. (2016) focus on the human work that goes into algorithm training and development, while Yang et al. (2021) emphasize the societal impact of AI. Moreover, for many researchers a key feature of HCAI is that AI should be transparent and explainable (Ehsan et al., 2021; Riedl, 2019; Sperrle et al., 2021). We propose that all of these conceptualizations of HCAI follow fundamentally from considering the human (broadly construed) as the main focus in AI development and differ largely as a function of the context in which this approach is applied.

Seeking to formalize these developments, a number of guidelines have been proposed by governments, organizations, and researchers to translate the ideals of HCAI into practice (Jobin et al., 2019). Many governments have proposed formal HCAI guidelines (Zhang et al., 2021). For example, the European Union lists seven key requirements that AI systems should meet in order to be trustworthy, including being transparent, having accountability, and promoting societal and environmental wellbeing (European Commission, 2019). Similarly, Australia has an AI ethics framework that includes principles such as fairness, human-centered values, and accountability (Australian Government, 2021), while China has released the ‘Beijing AI principles’ which include principles of doing good, being responsible, and being inclusive (Zhang et al., 2021). Among private companies, Microsoft has led the way in developing guidelines for ethical AI (Amershi et al., 2019; Madaio et al., 2020; Microsoft Microsoft Corporation, 2021). For example, Microsoft’s principles for ethical AI emphasize fairness, inclusiveness, reliability, safety, transparency, privacy, security, and accountability (Shaw, 2019). And finally, various researchers and research teams have proposed guidelines for AI. For example, Floridi et al. (2018) provide an ethical framework called AI4People that incorporates principles such as beneficence, justice, and explicability. Unfortunately, despite this proliferation of guidelines, the ideals of HCAI have proven difficult to put into practice (Oppermann et al., 2019; Shneiderman, 2020b; Winfield & Jirotko, 2018). Speaking to this point, Shneiderman (2021) has argued that while ethical guidelines are a step in the right direction, they are often too vague to be helpful for software engineers. Similarly, Mittelstadt (2019) has criticized AI ethics for consisting of vague principles and lofty value statements that lack the detail and precision needed to formulate specific recommendations for improving practice. Accordingly, it is not clear that HCAI in principle is reflected in the practices of AI developers.

1.2. AI user experiences

A broader issue with HCAI is that developing ever-more specific guidelines (e.g., the Microsoft AI fairness checklist) will not fully encompass the shift that is required to make AI truly human-centered. Rather, to focus on people, an HCAI approach must consider not only values but also needs, experiences, and impacts on organizations and society. One way to do this is to understand the experiences of people who use AI systems (Alvarado & Waern, 2018; Shneiderman, 2020b). Indeed, there is a growing literature on user experiences with AI systems, in which researchers have generally taken one of two approaches: (1) investigating a specific type of AI; or (2) investigating AI in a specific context. Examples of the former approach include user experiences with personal intelligence agents (Moussawi, 2018), conversational agents (X. Yang & Aurisicchio, 2021), and smart homes (Marikyan et al., 2019). The latter approach tends to focus on a particular group, such as medical practitioners (Maier et al., 2019), customers (Ameen et al., 2021), or novice users (Krening & Feigh, 2018). However, while informative for these specific domains, the results of these studies are less useful for assessing whether HCAI is sufficiently aligned with the needs of AI users.

This is because HCAI frameworks and guidelines generally aim to apply to all people, across all kinds of AI systems (Amershi et al., 2019; Huang et al., 2021; Jobin et al., 2019). In other words, it is not clear that the results of existing AI user-experience studies can be generalized to understand and assess HCAI more broadly.

1.3. The present study

HCAI currently consists of a collection of guidelines, frameworks, and principles, but there is a lack of empirical evidence regarding the extent to which these are utilized by AI developers, and whether HCAI in principle and in practice aligns with the needs of users. Accordingly, it is not clear to what extent HCAI can be considered to be truly ‘human-centered’ in the sense of fully understanding how the people who use AI systems perceive these and are affected by them. Moreover, whether or not user experiences are reflected in HCAI principles, theories, and guidelines, these high-level values may not be currently utilized by developers in ways which allow them to address the needs of users. In short, it is important for the field of HCAI to be aware of the extent to which it is connected with the people who make and use AI systems. In the present study we explored this topic empirically, aiming to address three key research questions:

RQ1. Do developer priorities align with user experiences?

RQ2. Do developer priorities align with HCAI in principle (i.e., theories, guidelines, and principles)?

RQ3. Does HCAI broadly construed (comprising HCAI in principle and developer priorities) align with user experiences?

2. Method

To address our research questions we engaged in a qualitative survey of AI users and developers. Responses to the survey were examined by means of a thematic content analysis, using a coding scheme that was created inductively based on patterns in responses, then iterated, and finally used to categorize all responses. From this, we were able to calculate the proportion of responses for each group that fit within a particular code (representing a theme in the dataset), allowing us to make comparisons between user and developers.

2.1. Participants

We used convenience sampling of several groups to capture a range of users and developers of AI systems. Specifically, a survey was distributed to 205 participants via online crowd-sourcing websites, university student research participation programs, and the researchers’ networks, with seven participant groups representing AI users ($N = 135$) and developers ($N = 70$). This sample size is commensurate with similar studies in the HCAI literature (e.g., Maier et al., 2019; Moussawi, 2018). Users included 50 US participants recruited from Prolific Academic ($M_{age} = 32$, 58% female), 75 undergraduate psychology students ($M_{age} = 20$, 77% female), and 10 postgraduate cyber security students ($M_{age} = 27$, 17% female). Developers included 12 tech professionals recruited via the research team’s networks ($M_{age} = 41$, 18% female), 36 undergraduate human-computer interaction (HCI) students ($M_{age} = 25$, 46% female), 18 undergraduate software engineering students ($M_{age} = 22$, 28% female), and 4 undergraduate machine learning students ($M_{age} = 22$, 50% female). All the student participants were recruited from an Australian university. The undergraduate psychology students received course credit for participation, while the other students did not receive compensation. Prolific Academic is an online platform that is often used to recruit participants for behavioral research (Palan & Schitter, 2018). The participants who were recruited via this platform were paid £2. The tech professionals did not receive compensation for participating. Together, these groups provide a cross-section of users and developers

from different countries, age groups, academic fields, and career stages.

To check whether the user group was comparatively less involved in AI development than the developer group, we asked participants whether they currently worked in tech, and whether they intended to in the future. Indicating that they were more involved in the development of tech than users, developers were twice as likely to say they worked in tech currently than users (31.4% versus 13.3%), and three times more likely to say they intend to work in tech in the future (74% versus 24.4%), indicating that they were more involved in the development of tech than users.

2.2. Survey items

The survey included the following open-ended questions: ‘What is a good experience you have had with AI? What made it good?’ (users only); ‘What is a bad experience you have had with AI? What made it bad?’ (users only); and ‘What is important to consider when designing AI?’ (developers only). The survey also included demographic measures, as well as number of other items that are not analyzed in the present paper.

2.3. Coding

Following data collection, participants’ responses were analyzed by the lead author, resulting in the creation of an initial set of 37 themes based on similarities across the dataset. Using this set of themes as codes, 30% of the dataset was coded by three members of the research team, with 10% coded by all coders and 20% coded uniquely. Inter-rater reliability was calculated using the 10% of data coded by all coders using Light’s kappa (Light, 1971), with the average inter-rater reliability being $\kappa = 0.34$. This represents a ‘fair’ (Landis & Koch, 1977) or ‘minimal’ (McHugh, 2012) level of reliability. Following this, the coders held a meeting to discuss discrepancies and refine the coding scheme, resulting in a revised list of 39 codes. From the remaining uncoded 70% of the dataset, the same three coders then coded an additional 10% with the new coding scheme. The average inter-reliability at this stage was $\kappa = 0.67$, representing a ‘substantial’ (Landis & Koch, 1977) or ‘moderate’ (McHugh, 2012) level of reliability. Given this acceptable level of inter-rater reliability, the lead author then recoded the full dataset in NVivo for Windows (version 12.6.1.970) using the revised coding scheme (see supplementary materials for full codebook). This approach was adapted from the ‘gold standard/master coder’ approach as described by Syed and Nelson (2015).

2.4. Comparisons

Percentages of AI users and developers whose responses were connected with each theme were calculated using NVivo and Microsoft Excel for Windows. Specifically, after the lead author recoded the full dataset in NVivo, we used NVivo’s coding comparison query function to calculate the number of responses corresponding to each theme for users and developers. We then created a Microsoft Excel spreadsheet to calculate the percentage of participants in each group whose response were associated with each theme. A copy of this spreadsheet, which includes the number of responses associated with each theme for each group, can be found in the supplementary materials.

3. Results

3.1. Developer perspectives and user experiences

Table 1 shows the top ten themes for each question together with the percentage of users and developers who referred to each theme. In the following paragraphs, we explain the most common themes (functionality and social impact), including examples of participants’ responses. We also make a number of key comparisons between developer

Table 1

Comparing top 10 themes for developer priorities with user experiences.

Developers (important considerations)	Users (good experience)	Users (bad experience)
Functionality (31%)	Social impact (42%)	Functionality (39%)
Ethics (20%)	Functionality (35%)	Unsure (21%)
Social impact (20%)	Unsure (18%)	Understanding user (16%)
Data (13%)	User experience (10%)	Social impact (13%)
Creation process (11%)	Understanding user (7%)	User-AI interaction (11%)
Privacy (9%)	Impressive (7%)	User experience (7%)
Efficiency (6%)	Easy to use (5%)	Emotion (6%)
Security (6%)	Safety (4%)	Privacy (5%)
Understandability (6%)	Efficiency (4%)	Ethics (4%)
Coding (4%)	Emotion (2%)	Understandability (4%)

priorities and user experiences with regards to these and other important themes. See the supplementary materials for the full set of themes, including explanations and interpretations.

3.1.1. Functionality

The code ‘functionality’ was applied to responses that mentioned functional elements of AI such as usability, usefulness, and performance. For the question ‘What is a good experience you have had with AI? What made it good?’, relevant responses included: “*Alexa. The algorithm adjusts based on your activities and what you like*”; “*Google maps, finding good routes with accurate timing predictions*.”; and “[a robot with AI] was good cause it worked really well and performed every task well”. For the question ‘What is a bad experience you have had with AI? What made it bad?’, responses included: “[In video games] the AI is often unpredictable, worse than a human opponent”; “*You Tube. Their algorithm insisting that I view political propaganda that I instructed them to stop showing me*.”; and “*I think the Gmail complete text function. It never guesses correctly*”. For the question ‘What is important to consider when designing AI?’, responses included: “*That it’s useful and accurate*”; “*No infinite loops*”; and “*How does it handle unexpected input outside the scope of its design and how generalisable is it*.”

Functionality was the defining theme of the dataset for both developers (31% of responses) and users (35% of responses for good AI experiences, 39% of responses for bad AI experiences). It was the most common theme among negative user experiences and the second most common theme among positive user experiences. This suggests that functionality is important for HCAI when it comes to creating good experiences but that it is even more important for avoiding bad user experiences (W. Xu, 2019). In other words, taking a ‘human-centered’ perspective means seeing functionality (as perceived by the user) as integral to the provision of utility, safety, and a positive user experience.

3.1.2. Social impact

‘Social impact’ was used to code references to the impact of AI on users, other people, and society. For the question ‘What is a good experience you have had with AI? What made it good?’, relevant responses included: “*I find Siri quite helpful. As a parent, I often have my hands full*.”; “*The traffic feature [on Google/Apple maps] allows me to avoid busy areas*”; and “*As a person who does not really like to call directly, [a customer service chatbot] helped a lot*”. For the question ‘What is a bad experience you have had with AI? What made it bad?’, responses included: “*Autocorrect ... It’s an example of AI making my life more inefficient*.”; “*When AI tries to learn who you are to serve advertisements and line pockets it feels like you’re being used, rather than being the user*”; and “*A bad experience I have had with AI is the YouTube algorithm and similar AI. It’s bad because these AI are really skilled at distracting me from my goals*”. For the question ‘What is important to consider when designing AI?’, responses included: “*is it interfering with my life or helping me?*”; “*Change management and how business/employees may react to it*”; and “*how AI will*

change others life”.

Relative to users, developers perceived social impact to be less of a priority when it comes to designing AI. This was particularly true for good (not just ‘not bad’) AI. Specifically, 20% of developers identified social impact as important, but it featured in 42% of positive user experiences. However, social impact was a feature of only 13% of negative user experiences. The particular importance of social impact for positive user experiences suggests that this is an aspect of AI that developers and guidelines should see as an integral part of HCAI.

3.1.3. Understandability versus understanding the user

A number of HCAI researchers have argued that AI should be understandable, explainable, and intelligible (Ehsan et al., 2021; Oppermann et al., 2019; Riedl, 2019; Sperrle et al., 2021). Indeed, this emphasis is widely reflected in ethical guidelines (European Commission, 2019; Floridi et al., 2018; Jobin et al., 2019; Shaw, 2019). However, our results suggest that making AI understandable is a low priority for developers (6%), and it was a negligible feature of user experiences (0% for positive experience, 4% for negative experiences). Instead, users were more concerned with AI understanding them, which was a feature of 7% of positive and 16% of negative user experiences. For example, in response to the question ‘What is a good experience you have had with AI? What made it good?’, users commented: “[AI] understands the purpose of human beings”; “it is personalised to me and my interests”; and “it helped provide me with the correct information I wanted”. Similarly, in answer to the question ‘What is a bad experience you have had with AI? What made it bad?’, users remarked: “When they [chatbots] aren’t programmed correctly, it’s like holding a conversation with someone who isn’t listening or talking to you directly.”; “Sometimes because it is controlled by technology, they are not emotionally sensitive and at times you need that.”; and “Siri not being able to understand ethnic names and calling the wrong people”.

3.1.4. Ethics, privacy and security

Developers considered ethics (20%), privacy (11%), and security (6%) to be important in designing AI. Promisingly, this accords with the emphasis of ethical AI guidelines (Jobin et al., 2019). However, these themes provide a clear contrast between the priorities of developers and those identified through the experiences of users—who only mentioned privacy (5%) and ethics (4%) occasionally with relation to negative experiences. Examples of developer responses that referred to ethics included: “A few AI have been found to have racial bias which is a great concern.”; “What ‘values’ can be baked into AI that represent human values?”; and “how does AI work on a legal way?”. Some developers also mentioned privacy specifically, commenting that: “it could be a problem if AI can access our web browser or purchase records to do something else such as targeted advertising ...”; “It should not be able to intrude on the privacy of users unless allowed to.”; and “If the data can’t be depersonalised, it must have a method of being completely removed without trace if the user requests it.”. Developers occasionally referred to security, although without elaboration, e.g., “Security and privacy are also important”; “Human values and security”; and “Security, privacy and ethics”.

3.1.5. Unsure

Users were often unable to think of a good or bad experience with AI. Indeed, ‘unsure’ was the third most common theme for good experiences (accounting for 18% of user responses) and the second most common theme for bad experiences (accounting for 21% of user responses). There may be a range of reasons for this, including the open-ended nature of the questions, the broad terms in which these questions were phrased, a genuine lack of experiences with AI systems, or a lack of user knowledge regarding what constitutes AI and where they are interacting with it (Curtis et al., 2022; Gillespie et al., 2021). To explore this further, we summarized the kinds of AI systems that users identified. In general, these were legitimate examples of AI, including *humanoid AI* (Alexa, Siri, Cortana, chatbots, computer game AI, AI teachers, AI server at

restaurant), *non-humanoid AI assistance* (driving assistants, health AI, AI for spotting anomalies in data-rich environments, Google Home), *algorithmic recommendations* (YouTube, TikTok, Instagram, online shopping suggestions, Spotify playlists), *search processes* (Google search, Google reverse image search), *predictive AI* (Gmail email prediction, Google search predictions, Microsoft Office autocorrect), *AI underlying the functionality of other apps or software* (ride-hailing applications, face detection and recognition, music conversion app, voice recognition, fingerprint scanning), and *AI in hardware* (Roomba, rescue robots, Tesla). However, there were a few examples of systems that do not, or may not have, an AI component, such as calculators, self-serve check-outs, ATMs, and virtual reality systems.

4. Discussion

We investigated user experiences and developer priorities to explore whether HCAI (in principle and practice) is truly human-centered. Our key findings and implications for HCAI are summarized in Table 2. Developer priorities generally aligned with HCAI guidelines. For example, developers considered ethics, privacy, security, and understandability as important to consider when designing AI, all of which feature prominently in existing guidelines (Jobin et al., 2019). Given that many participants in the developer samples were university students, it is encouraging that emerging developers appear to have internalized these important values. Developers also aligned with existing guidelines in that they prioritized the performance of AI systems but went further by focusing on other aspects of functionality such as usefulness and usability. These were defining features of both positive and negative user experiences. In this sense, developers have an intuitive sense that a key element of making AI human-centered is to ensure that it works well for the user. While researchers have noted the importance of this expanded conceptualization of functionality in HCAI (Shneiderman, 2021; W. Xu, 2019), it is critical that HCAI guidelines further emphasize this point, not least so that they seem intuitive and grounded in reality to developers. Our results suggest that functionality from the perspective of the user is crucial for HCAI.

Furthermore, our results suggest that the impact of AI on users, other people, and society is a defining feature of user experiences, particularly positive experiences. However, while developers did prioritize social impact, they did so to a lesser extent than users. Interestingly, developers saw social impact and ethics as equally important. This aligns with existing HCAI guidelines, which tend to conceptualize social impact in terms of broad values and society-level impacts (Jobin et al., 2019; Mittelstadt, 2019; Shneiderman, 2021). However, while ethics and values are important to consider in relation to AI, users only rarely

Table 2
Key Findings and their Implications for HCAI.

Key findings	Implications for HCAI
➤ Functionality was a defining feature of user experiences with AI and was a key priority for developers	• Making AI systems work for the user is a vital part of making them human-centered
➤ Social impact was a defining feature of user experiences with AI, particularly positive experiences	• HCAI requires a greater focus on, and clearer conceptualization of, the impact of AI on users and other people.
➤ Understandability was less important for users than being understood by AI	• While understandability, explainability, and interpretability are important for HCAI, it is also important that AI systems make the user feel understood
➤ Ethics, privacy, and security were important developer priorities, but were not key aspects of AI user experiences	• AI policies tend to reflect a focus on the avoidance of harm, which is essential but on its own not sufficient to fulfill the goals of HCAI
➤ Many users could not identify an example of a positive or negative experience with AI	• More research is required to understand the extent to which people are aware of AI systems in their lives

mentioned ethics or values in their experiences. Rather, their experiences were shaped by the impact of AI on themselves and others. On this basis, we suggest that more needs to be done to recognize and conceptualize the social impacts of AI, particularly when seeking to understand positive user experiences and their translation of this into developer practices. Similarly, it may be that HCAI guidelines could benefit from an increased focus on understanding the needs and wants of people. Our results also reinforce the idea that participatory design with potential users and other stakeholders is important for AI to be human-centered (Axelsson et al., 2021; Elahi et al., 2021; Shneiderman, 2020b).

HCAI researchers and guidelines emphasize the idea that AI should be explainable, understandable, and interpretable to researchers, users, and those affected by AI system (Ehsan et al., 2021; European Commission, 2019; Floridi et al., 2018; Jobin et al., 2019; Oppermann et al., 2019; Riedl, 2019; Shaw, 2019; Sperrle et al., 2021). However, this focus may obscure the importance of users feeling *understood* by AI (Riedl, 2019). Indeed, users in our study did not seem concerned about understanding AI but were often frustrated by a perceived lack of understanding on the part of AI—whether that was understanding them as a person, understanding what they wanted from the AI system, or understanding humans in general. These results align with Bucher's (2017) findings that negative experiences such as feelings of being profiled, misunderstood, or ignored can follow from AI systems being unable to understand people in general, and the user in particular. Solutions to such negative experiences have also been proposed by a number of researchers (Alvarado & Waern, 2018; Eslami et al., 2015; Nagulendra & Vassileva, 2014, pp. 107–115). However, these have generally focused more on transparency and user control than on improving the extent to which AI systems and the people who create and implement them understand humans. While user control and understanding are important components of allowing AI systems to understand the user as a unique individual, we propose that it is also important for AI systems and the people who create and implement them to better understand people in general and incorporate these understandings to improve the 'algorithmic experience' (Alvarado & Waern, 2018).

In our study, developers prioritized ethics, privacy, and security. In other words, they showed a keen awareness of the dangers of AI systems. However, the importance of social impact in defining positive user experiences demonstrates that another key element of HCAI is that it should make people's lives *better* (Floridi et al., 2018; Shneiderman, 2021; W. Xu, 2019). In a review of recent AI guidelines, Jobin et al. (2019) found that transparency, justice, fairness, non-maleficence, responsibility, and privacy were all more commonly mentioned than beneficence. This reflects a general focus in existing principles towards the prevention of harm. This is critical but should not be the sole goal of HCAI. For example, Floridi et al. (2018) have proposed that AI might be used to enable self-realization, enhance agency, increase societal capabilities, and cultivate societal cohesion. However, these outcomes cannot be achieved by focusing purely on avoiding or reducing harm. Accordingly, it is important to also understand what users want and need, and to incorporate this into the development process on an equal footing alongside non-maleficence. An example of how HCAI can focus on understanding and fulfilling the needs of the user while still minimizing negative outcomes is provided by Loi (2019), who sets out ten guidelines for intelligent systems that include 'Make people feel unique and empower their unique goals', 'Do not make systems human, but capable of helping humans', and 'Prioritize usages that matter – helper usages'. Similarly, Bellet et al. (2021) used a user experience methodology when designing an AI algorithm to manage human-machine transitions with vehicle automation, including measures of utility, satisfaction, and safety.

4.1. Incorporating human needs into HCAI

The findings of the present study indicate that **social impact** is a major **concern for both developers and users of AI systems**. However,

relying solely on ethical frameworks to increase positive and reduce negative impacts is unlikely to be effective (Eitel-Porter, 2021), while checklists concerning specific practices have the potential to become 'box ticking' exercises rather than encouraging a human-centered focus (Bosk et al., 2009; Kiran et al., 2015). Moreover, it is currently unclear what might constitute a positive social impact with regards to AI (Floridi et al., 2018; Moore, 2019). We suggest that an effective way to conceptualize social impact from a human-centered perspective is to focus on universal human needs. Specifically, there is a large body of research demonstrating that fulfillment of psychological needs is important for individual outcomes such as physical health, psychological wellbeing, and flourishing, as well as for group outcomes such as motivation, communication, and productivity (Deci & Ryan, 2012; S. A. Haslam, 2004; Jetten et al., 2012). Indeed, along these lines, several researchers have discussed the importance for HCAI of understanding human needs (Floridi et al., 2018; W. Xu, 2019). Moreover, other HCAI researchers have investigated the needs of specific groups, such as medical practitioners (Cai et al., 2019), people who need mobile translation (Liebling et al., 2020), and developers (Holstein et al., 2019). However, with regards to specific needs, HCAI has focused mainly on autonomy (Floridi et al., 2018; Maedche et al., 2019; Väänänen et al., 2020).

Drawing on research in social psychology (Baumeister & Leary, 1995; Deci & Ryan, 2012; Tajfel & Turner, 1979; Turner et al., 1987, p. 239) we propose that one way to ensure that HCAI works 'for good' is to focus on the specific needs that people seek to satisfy in order to thrive, with the aim of incorporating this understanding into development, implementation and evaluation cycles to ensure that AI systems facilitate these needs. As a starting point, we argue that it is vital that AI enables people to strengthen their connections to groups and society in psychologically meaningful ways (C. Haslam et al., 2018; Turner et al., 1987, p. 239). Moreover, while HCAI should aim to maximize the autonomy of users, it should also consider the ability of groups to achieve shared goals (particularly those which are marginalized in broader society (Ho, 2020; Pal et al., 2018; Trewin et al., 2019)). Furthermore, AI systems should support both users and the groups they belong to feel competent (Deci & Ryan, 2012; Kachanoff et al., 2020). By analyzing the social impact of AI at the social psychological level of analysis (i.e., its effects on people and the groups they belong to), the meaning of 'positive impact' can be operationalized by making use of existing theorizing and tools for measurement (Baumeister & Leary, 1995; Deci & Ryan, 2012; Ng et al., 2012; Postmes et al., 2013; Tajfel & Turner, 1979; Turner et al., 1987, p. 239). In this way, AI 'for good' can be described in ways that differentiate it from AI 'for bad' (i.e., AI which has a negative impact on people).

4.2. Strengths, limitations, and future research

The specific sample of people surveyed in this study represents both a strength and a limitation. We surveyed users from a variety of backgrounds across two countries, tech industry professionals, and emerging developers attending university courses. Accordingly, the study provides a different perspective to previous AI user experience studies, which have tended to focus on a particular group (Ameen et al., 2021; Krening & Feigh, 2018; Maier et al., 2019; S. J. H. Yang et al., 2021). Moreover, the inclusion of developers 'in training' allowed us to gain a snapshot not only of priorities that are informed by current practice, but also of cutting-edge thinking in HCAI. Indeed, approximately half of these students were recruited from an HCI university course in which HCAI theories and guidelines are explicitly taught. However, our sample is less representative of developers who have been working in AI for many years. Future research should survey a larger structured sample of established industry professionals to see whether the results we have observed differ amongst developers who have more varied experience with AI in practice. Given representative surveys show that younger generations are more trusting and accepting of AI systems than older

generations (Gillespie et al., 2021), future research should examine whether the findings we report hold for older people. It is also important for future research to incorporate a broader range of user perspectives, noting that our sample was both young and quite positive about AI in general. In particular, it is important to capture the experiences of people in marginalized and vulnerable groups. For example, users who experience AI systems as perpetuating existing structural inequalities may have a different conceptualization of ‘social impact’ that it is important to identify and acknowledge. To improve the sampling procedure, future research should also incorporate a more iterative process of data collection and analysis to ensure data saturation. Future research might also explore how the ethical values of HCAI can be translated into specific practices — for example, by adopting a needs-based approach.

Another aspect of our study that is both a strength and limitation is the set of questions that we asked participants. These were simple and broad, and for this reason the answers we received can be compared against the generally broad scope of existing HCAI theorizing and guidelines. This contrasts with existing user experience research in HCAI, which has tended to focus on specific AI systems (Marikyan et al., 2019; Moussawi, 2018; Y. Xu & Warschauer, 2020; S. J. H. Yang et al., 2021). However, future research in this area should develop more targeted questions to build on these initial findings. For example, some of the major themes identified in our analysis could guide the creation of specific questions about functionality, personalization, and human needs. To generate richer responses, these questions could also be posed in semi-structured interviews. Moreover, many users were not able to think of a positive or negative experience with AI. While this may be a function of the open-ended nature of the questions, it could also be because these participants lacked a clear understanding of what constitutes AI. Future research should provide users with a definition of AI or examples of AI-enabled systems to help clarify this for participants.

Finally, there are a number of intriguing questions raised by our results which are worth exploring in future research. For example, users spontaneously identified a number of examples of AI across a wide variety of domains. There is a long tradition of defining AI from a theoretical or expert perspective (Marr, 1977; McCarthy, 2004; Monett & Lewis, 2018), but given the prevalence of AI systems in people’s everyday lives it is important to establish what non-experts think AI is, as this is likely to affect how they feel about the extent to which it should be implemented or regulated (Walsh, 2022). Moreover, although our data suggest that developer priorities are aligned with at least some elements of ethical frameworks, it is important to explore whether developers actively make use of these frameworks. Our results suggest that frameworks may be effective in orienting developers to values that are widely considered to be important, but they may lack utility when it comes to making specific design choices.

5. Conclusion

In this paper we reviewed HCAI theory and guidelines and compared these with the results of a survey assessing developer priorities and user experiences. Developer priorities were aligned to some extent with current guidelines, although our results suggest that AI both in principle and in practice falls short of the objectives set out by HCAI theory. Promisingly, developers were aware of the risks of AI systems, and beyond this had an intuitive understanding that improving functionality from the perspective of the user is an important goal for HCAI. However, our results suggest that HCAI stands to benefit from an increased understanding of the social impacts of AI, as these are particularly important for positive user experiences and it is these that may ultimately further shape the AI ethics landscape. Furthermore, to bridge the gap between the goals of HCAI and current practice, we propose that researchers and guidelines should focus more on understanding what people need in their lives and how AI helps or hinders the satisfaction of these needs. Ultimately then, we would suggest that to be truly ‘human-centered’ HCAI needs to focus much more on humans than it currently

does.

Credit author statement

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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Appendix A. Supplementary data

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